

BUY, Russian Federation

WMO: 272420

Lat: 58.483N

Lon: 41.533E

Elev: 338

StdP: 14.52

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-13.9	-25.1	1.4	-19.8	-18.8	2.0	-13.6	13.5	16.7	11.9	19.5	1.7	30	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	85.3	69.0	81.4	67.6	77.9	65.9	71.3	81.4	69.3	78.6	67.4	75.6	4.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.9	103.7	76.2	65.9	96.9	74.6	64.1	90.7	72.3	35.3	81.4	33.6	78.6	32.0	75.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.0	9.1	DB	-26.8	89.7	6.6	4.2	-31.5	92.7	-35.4	95.2	-39.1	97.6	-43.9	100.7	(4)
(5)			WB	-27.0	74.2	6.6	2.2	-31.7	75.8	-35.6	77.1	-39.3	78.3	-44.0	79.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.4	13.5	15.6	25.6	40.3	52.6	60.7	66.1	60.7	50.7	39.4	27.2	18.9		
	DBStd	20.54	13.87	13.46	8.98	8.79	8.98	7.27	6.46	6.68	7.21	7.91	10.21	13.52		
	HDD50	5312	1130	964	758	312	77	5	0	3	77	338	683	965		
	HDD65	9526	1595	1384	1223	740	395	169	64	169	431	793	1133	1430		
	CDD50	1447	0	0	0	22	158	325	501	333	98	10	0	0		
	CDD65	186	0	0	0	0	11	40	99	34	2	0	0	0		
	CDH74	1758	0	0	0	5	119	414	904	303	13	0	0	0		
	CDH80	470	0	0	0	0	29	108	247	85	1	0	0	0		
	WSAvg	3.8	4.1	4.2	4.2	3.8	4.0	3.4	2.9	3.1	3.3	4.0	4.3	4.4		
	PrecAvg	24.2	1.7	1.4	1.4	1.3	1.9	2.8	2.9	2.6	2.1	2.5	2.0	1.7		
(15) Precipitation	PrecMax	33.1	3.0	2.7	2.8	2.9	5.1	5.4	6.6	8.2	4.2	5.3	3.7	3.0		
	PrecMin	18.3	0.8	0.2	0.1	0.6	0.5	0.7	0.0	0.3	0.9	0.9	0.4	0.7		
	PrecStd	3.8	0.6	0.7	0.7	0.6	1.0	1.3	1.8	1.7	0.9	1.2	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	37.6	51.2	72.8	84.3	87.9	90.8	88.1	75.4	63.7	49.0	42.7		
		MCWB	34.7	34.7	41.6	56.5	66.7	70.2	71.5	70.6	63.6	55.3	47.5	41.5		
	2%	DB	34.2	35.3	43.8	66.4	77.2	83.3	85.6	81.0	70.0	57.2	44.3	38.3		
		MCWB	33.1	33.8	37.3	53.1	62.9	67.6	69.5	67.8	61.1	51.7	42.7	36.8		
	5%	DB	33.1	34.0	39.9	60.7	72.6	79.0	82.3	76.8	65.6	53.0	41.5	35.3		
		MCWB	32.2	32.8	35.1	50.0	58.4	65.9	68.5	65.6	58.3	49.3	40.3	34.1		
	10%	DB	31.7	32.7	37.2	55.4	68.7	74.8	79.0	73.0	62.0	50.1	38.8	33.6		
		MCWB	30.8	31.7	33.9	46.2	56.1	63.3	67.2	64.0	56.4	47.5	37.6	32.7		
	0.4%	WB	35.1	35.6	42.6	57.7	69.8	73.1	74.9	73.5	67.0	56.5	48.0	41.5		
		MCDB	35.6	36.6	49.4	70.3	81.5	83.7	84.9	84.4	72.0	61.7	49.4	42.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.5	34.1	38.2	54.1	64.5	69.7	71.9	69.7	62.5	52.6	43.1	37.0		
		MCDB	33.9	34.9	42.1	64.9	73.8	80.4	82.2	78.1	67.7	56.0	43.9	37.8		
	5%	WB	32.4	33.0	36.0	50.7	60.9	67.2	70.0	67.4	59.5	49.7	40.5	34.3		
		MCDB	33.1	33.8	38.8	59.3	69.6	76.3	79.5	74.6	64.2	52.2	41.4	34.9		
	10%	WB	30.7	31.7	34.4	47.4	57.9	64.7	68.2	64.8	56.7	47.6	37.6	32.8		
		MCDB	31.7	32.7	36.7	54.5	65.9	72.5	76.5	71.8	61.4	50.1	38.5	33.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	13.2	15.6	17.7	19.4	18.5	18.7	17.2	14.7	9.7	7.2	8.6		
		MCDBR	7.4	8.7	18.3	26.0	24.8	24.2	23.2	23.2	20.6	15.2	8.6	5.8		
		MCWBR	7.4	8.1	12.4	14.8	12.0	10.9	9.9	11.3	12.0	10.1	7.9	5.9		
		MCDBR	7.2	8.3	13.6	24.2	21.2	21.3	20.6	20.5	17.6	13.4	7.9	5.7		
	5% WB	MCWBR	7.3	8.1	9.7	14.2	12.0	10.6	10.2	10.8	11.1	9.8	7.7	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.316	0.345	0.396	0.372	0.367	0.404	0.400	0.363	0.342	0.306	0.266		
	taud		2.066	2.056	2.058	2.132	2.378	2.410	2.329	2.317	2.404	2.345	2.178	2.044		
	Ebn at Noon		165	215	246	254	273	276	261	252	245	211	155	127		
	Edh at Noon		17	28	37	41	34	34	36	34	27	21	15	12		
(44) All-Sky Solar Radiation	RadAvg		129	368	817	1251	1673	1751	1728	1325	769	333	114	70		
	RadStd		21	72	88	140	129	144	175	94	98	56	19	15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page